

JP Seismic Tomography - Requirements and Restrictions

Computational resources, benchmark conditions and changes that affect reproducibility

1. Purpose of this document

This document records the practical conditions that matter when reproducing the JP seismic-tomography benchmark. Some conditions are mathematical and should remain fixed because changing them changes the experiment. Others are computing-resource conditions that depend on the University of Manchester CSF allocation available to the person carrying out the reproduction.

2. Matrix and dataset requirements

The benchmark is tied to the SuiteSparse Matrix Collection dataset identified in the code as `Harvard_Seismology/JP`. The expected matrix has 87,616 rows, 67,320 columns and 13,734,559 non-zero entries. It is stored and processed as a sparse matrix.

A different matrix, a dense conversion of the matrix, or a different version of the dataset should not be substituted without explicitly describing the change. A successful run on another matrix is not a reproduction of this experiment.

3. Right-hand side restriction

The benchmark does not use an original observed seismic right-hand side. The numerical program constructs the vector using a reference vector of ones:

$$b = A @ \text{ones}$$

This construction must be preserved. Replacing b with another vector, including a real-world observational vector, creates a different experiment. The controlled right-hand-side construction is intentional because the purpose here is to compare solvers on the real JP operator under reproducible conditions.

4. Mathematical formulation

For each λ , the benchmark solves the ridge-regularised least-squares problem

$$\text{minimise over } x: \|Ax - b\|_2^2 + \lambda \|x\|_2^2$$

CG and PCG use the regularised normal-equation operator without explicitly storing $A^T A$. PCG adds a Jacobi diagonal preconditioner. LSQR and LSMR use the sparse matrix directly in their damped least-squares formulation. This distinction is part of the implementation being benchmarked and should not be removed merely to make the code easier to understand.

5. Solver set

Method	Restriction
CG	Retain the matrix-free normal-equation implementation.
PCG	Retain the Jacobi diagonal preconditioner and matrix-free normal-equation implementation.

Method	Restriction
LSQR	Use the supplied sparse least-squares implementation rather than replacing it with a direct factorisation.
LSMR	Use the supplied sparse least-squares implementation and preserve the damping relationship with lambda.

6. Regularisation values

The dissertation's JP results used the explicit sweep 1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1, 10. The current repository script also contains an adaptive default mode. For a strict reproduction of the dissertation figures and reported CSV results, use the explicit eight-value sweep rather than changing the lambda range for convenience.

Lambda is an experimental parameter. Choosing a single value because it happens to give the fastest result would remove the comparison that the experiment was designed to make.

7. Tolerance, iteration limit and repeats

The current supplied JP_Seismic_Tomography.py defines $RTOL = 1e-8$, $MAXITER = 10000$ and $DEFAULT_REPEATS = 3$. The archived dissertation output shows the JP benchmark being run with a 5000-iteration limit. This difference matters if the goal is to reproduce the archived numerical results exactly. The code version used for a reproduction should therefore be recorded, and the iteration limit should not be silently changed.

In particular, a run reaching the iteration limit is meaningful information. It should not be hidden by increasing MAXITER after the fact simply to make the solver appear to have converged.

8. CSF and Slurm requirements

The large JP benchmark was run on the University of Manchester CSF using Slurm. A researcher reproducing it must have legitimate CSF access and must follow the current University procedure for logging in, selecting an environment and submitting jobs. Those login instructions are intentionally not fixed here because they are University procedures rather than properties of the experiment.

The dissertation execution used a 24-hour wall-time allocation. A 24-hour request should therefore be regarded as the documented benchmark allocation, subject to the current CSF policy and available resources.

9. Memory requirement

Memory availability is especially important for this experiment because the JP matrix is large and the normal-equation calculations involve substantial intermediate vectors and sparse operations. The practical allocation should be checked before submission.

For the dissertation-scale direct approaches used elsewhere in the project, allocations below 128 GB RAM should be treated as insufficient for reliable reproduction. For the JP benchmark itself, the iterative methods are designed to avoid explicitly forming $A^T A$, but the workload is still large and memory use can be substantial. A larger memory allocation does not guarantee that every method will finish successfully.

The University allocation available to a particular user must therefore be checked before running the experiment. The current CSF documentation should be consulted for the maximum memory request, permitted wall time, available partitions and any account-specific limits.

10. CPU allocation

The number of CPUs available to a Slurm job is user- and policy-dependent. It must be checked against the current University allocation before the job is submitted. The reproduction documents should record the CPU count actually requested and, where possible, the CPU configuration reported by Slurm.

The CPU request should not be treated as a universal benchmark constant. Changing the number of CPUs can change runtime measurements, particularly when numerical libraries use threaded low-level operations.

11. Direct solvers and feasibility

Direct solution methods are not equivalent to the matrix-free iterative implementations used for the main JP comparison. A direct approach can require explicit formation or factorisation of large intermediate matrices and can therefore become infeasible even when substantial memory is available. Memory capacity alone should not be interpreted as a guarantee that a direct solver will complete.

For this reason, the JP benchmark should not be altered by introducing an explicit A-transpose-A matrix merely to make CG or PCG easier to implement. The matrix-free operator is part of the experiment.

12. Timing restrictions

- Runtime measurements should be taken from the numerical program's own timing region.
- Do not mix queue waiting time with solver runtime.
- Do not compare timings from substantially different CPU or memory allocations without stating the difference.
- Do not report a single run as if it were the three-repeat benchmark.
- Keep the generated CSVs so that the means and variability can be inspected after the run.

13. Plotting restrictions

plot_JP.py is a post-processing step. It should read the CSV outputs produced by the numerical benchmark. It should not be edited to manufacture missing values or to replace failed solver runs with manually entered numbers. If a CSV is incomplete, the numerical job should be investigated first.

14. Changes that invalidate a strict reproduction

- Replacing the JP matrix with another dataset.
- Changing b from $A @ \text{ones}$ to a different right-hand side.
- Removing one or more solvers from the comparison.
- Changing the lambda sweep without clearly labelling the new experiment.
- Changing tolerance or iteration limits without recording the change.
- Replacing matrix-free normal equations with an explicitly stored A-transpose-A matrix.
- Replacing PCG's Jacobi preconditioner with a different preconditioner.
- Changing the number of repeats and still describing the result as the original benchmark.

- Treating a different CSF hardware or resource allocation as though it were identical to the dissertation execution.

15. What should be recorded for a reproducible rerun

- Date of the run.
- Version or commit of the repository code.
- Python version and relevant NumPy, SciPy, pandas and Matplotlib versions.
- JP matrix source and matrix dimensions/non-zero count.
- Lambda values used.
- Tolerance and maximum iteration limit.
- Number of repeats.
- Slurm wall-time request.
- Slurm memory request and actual maximum resident memory where available.
- CPU allocation.
- Any changes made to the supplied scripts or environment.

16. Generated files do not need to remain in the repository

The CSV files and PNG plots are reproducible outputs rather than source material. It is reasonable for the repository folder to contain only the Python programs and the three explanatory PDFs. A researcher can regenerate the CSV and PNG files by following the Run Instructions document. If generated outputs are retained elsewhere, their location and provenance should be recorded.

17. Final reproducibility rule

The safest rule is simple: preserve the supplied code and record any unavoidable change. Resource requests such as memory, CPUs, partition and wall time are necessarily dependent on the current University environment; mathematical inputs, solver definitions and benchmark parameters are part of the experiment itself. Keeping those two categories separate makes the work reproducible without pretending that a future CSF allocation must be identical to an old one.